

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 4 MODULE, AUTUMN SEMESTER 2014-2015

COMPUTER VISION

Time allowed TWO HOURS

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer THREE Questions

Each question carries 33 marks

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn your examination paper over until instructed to do so

ADDITIONAL MATERIAL: None

INFORMATION FOR INVIGILATORS: None

1. Optic Flow

(a) In the context of optical flow, explain what is meant by the terms

- i) Brightness constancy
- ii) Smoothness

(6 marks)

(b) What smoothness measure is used in the Horn-Schunck algorithm?

(3 marks)

(c) Horn and Schunck proposed the following two equations for computing optical flow in an iterative manner:

$$u_t = \bar{u} - \lambda \frac{\frac{\partial I}{\partial x} \bar{u} + \bar{v} \frac{\partial I}{\partial y} + \frac{\partial I}{\partial t}}{1 - \lambda \left(\left(\frac{\partial I}{\partial x} \right)^2 + \left(\frac{\partial I}{\partial y} \right)^2 \right)}$$

$$v_t = \bar{v} - \lambda \frac{\bar{u} \frac{\partial I}{\partial x} + \bar{v} \frac{\partial I}{\partial y} + \frac{\partial I}{\partial t}}{1 - \lambda \left(\left(\frac{\partial I}{\partial x} \right)^2 + \left(\frac{\partial I}{\partial y} \right)^2 \right)}.$$

Discuss the role of λ in these equations.

(3 marks)

(d) These equations suffer from a 'chicken and egg' problem where in order to compute u and v we must first compute $\bar{u}$ and $\bar{v}$, which in turn depend on u and v . How this problem is resolved in practice?

(6 marks)

(e) Describe how an *image pyramid* can be used in association with the Horn-Schunck algorithm, and what advantages a pyramid-based scheme would give.

(7 marks)

(f) Figure 1 was obtained from a traffic monitoring camera which acquires images at 25 frames/second. The truck is driving away from the camera, the tree is swaying in the breeze. What problems would arise if the Horn-Schunck algorithm was used to extract motion information from a pair of neighbouring images from this sequence? How might these problems be addressed?

(8 marks)



Figure 1

2. Feature Detection and Segmentation

(a) Image features are central to successful object recognition and tracking. Briefly describe the four basic features employed in the Viola-Jones object recognition algorithm. (6 marks)

(b) The Viola-Jones features can be computed efficiently due to Viola and Jones' introduction of the *integral image*.

(i) What is the integral image, and how is it computed? (5 marks)

(ii) Given an integral image, how would you calculate the sum of the intensity values present in a 10 x 15 image patch whose top-left corner lies at pixel (20, 20)? (3 marks)

(c) Snakes identify region boundaries by minimising an energy function along control points that define a curve. This function usually has two components – an internal energy E_{int} and an external energy E_{ext} . The two terms are combined to give the total energy $E_{\text{snake}} = \alpha E_{\text{int}} + \beta E_{\text{ext}}$.

(i) What is the difference between the internal and external energy components? (3 marks)

(ii) E_{ext} is often related to the image gradient, with

$$E_{\text{ext}} = -\sum_i \left(\left(\frac{\partial I}{\partial x} \right)^2 + \left(\frac{\partial I}{\partial y} \right)^2 \right)$$

being widely used. Why is this term negated?

(3 marks)

(iii) What requirement does the internal energy term $E_{\text{curve}} = |p_{i-1} - 2p_i + p_{i+1}|^2$ place on the contour?

(3 marks)

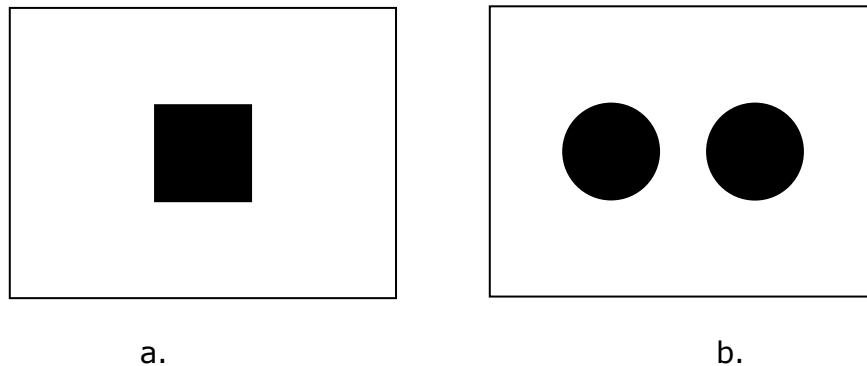


Figure 2

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Turn Over

(iv) A snake employing the energy terms E_{ext} and E_{curve} given above and designed to shrink towards its target is initialised near the boundary of the image sketched in Figure 2a. Briefly describe the final shape of the snake, and explain why that shape will appear.

(5 marks)

(v) A snake employing the energy functions given above and designed to shrink towards its target is initialised near the boundary of the image sketched in Figure 2b. What is the effect on the shape of the final snake of setting α much greater than β , and β much greater than α , respectively?

(5 marks)

3. Tracking

Condensation tracks objects by maintaining a set of hypothesised states. At each time point this set of states is sampled to make a set of predictions, the predictions are affected by random noise, and then evaluated against the observed image.

(a) Briefly explain what is involved in each of the *prediction*, *noise*, and *evaluation* steps, and what the purpose of each of these steps is.

(12 marks)

(b) An advantage of Condensation over other methods is that it is able to cope with *multi-modal* distributions.

(i) What does it mean for a distribution to be multi-modal?

(ii) Why is Condensation robust to such distributions while techniques such as Mean Shift are not?

(iii) Why is the ability to cope with multi-modal distributions important in a visual tracker?

(8 marks)

(b) You are building a vision-based interface, and to close the application you need to track the hand of a person facing a colour camera and waving goodbye. Assuming that you use a particle filtering approach, what problems might you face, and how might they be addressed?

(13 marks)

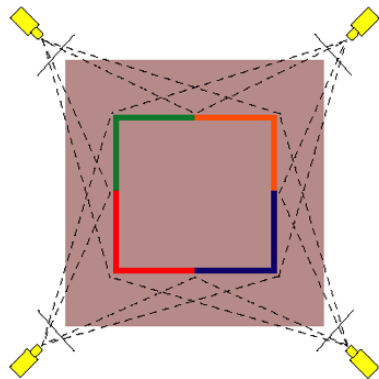
4. Stereo Vision

(a) Explain with the aid of a diagram what is meant by the term *epipolar plane*, and how knowledge of the epipolar plane simplifies the stereo correspondence problem. (8 marks)

(b) Epipolar geometry was first used in binocular stereo, but is also valuable when recovering depth from multiple views. What are the advantages and disadvantages associated with using more than two views? (6 marks)

(c) Multiple baseline stereo was among the first attempts to generalise binocular stereo methods to multiple views. Describe the operation of the multiple baseline algorithm, and briefly discuss its particular strengths and weaknesses. (11 marks)

(d) More recent multi-view stereo methods employ less restrictive camera geometries. The figure below shows the top view of four cameras imaging a coloured rectangular solid.



With reference to this situation, explain what is meant by the terms *photo hull* and *visual hull*. (8 marks)